Federated electronic health records for the European Health Data Space

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The European Commission's draft for the European Health Data Space (EHDS) aims to empower citizens to access their personal health data and share it with physicians and other health-care providers. It further defines procedures for the secondary use of electronic health data for research and development. Although this planned legislation is undoubtedly a step in the right direction, implementation approaches could potentially result in centralised data silos that pose data privacy and security risks for individuals. To address this concern, we propose federated personal health data spaces, a novel architecture for storing, managing, and sharing personal electronic health records that puts citizens at the centre—both conceptually and technologically. The proposed architecture puts citizens in control by storing personal health data on a combination of personal devices rather than in centralised data silos. We describe how this federated architecture fits within the EHDS and can enable the same features as centralised systems while protecting the privacy of citizens. We further argue that increased privacy and control do not contradict the use of electronic health data for research and development. Instead, data sovereignty and transparency encourage active participation in studies and data sharing. This combination of privacy-by-design and transparent, privacy-preserving data sharing can enable health-care leaders to break the privacy-exploitation barrier, which currently limits the secondary use of health data in many cases.

Lancet Digit Health 2023; 5: e840–47

Published Online September 21, 2023 https://doi.org/10.1016/S2589-7500(23)00156-5

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Introduction

The application of artificial intelligence (AI) to medicine has resulted in substantial improvements in many fields; for example, in the diagnosis of cancer[1] and drug discovery.[2] However, AI systems driven by training data require large quantities of curated and annotated medical data.

Compiling adequate research databases outside of dedicated studies requires standardised and unified digital medical data storage on a national or even international scale and legal frameworks regulating and governing data access. Efforts to establish national research databases in several countries have been ongoing for decades (eg, Clinical Practice Research Datalink and Optimum Patient Care Research Database in the UK,[3,4] Medizininformatik-Initiative in Germany,[5] Sundhedsdatastyrelsen in Denmark,[6] and Estonian National Health Information System in Estonia[7]). The COVID-19 pandemic further fuelled the development and implementation of digital health solutions overall and specifically for tracking the COVID-19 pandemic and vaccine effectiveness using health data (eg, EAVE II in Scotland,[8] OpenSAFELY in the UK,[9] and Netzwerk Universitätsmedizin COVID-19 Data Exchange Platform and Netzwerk Universitätsmedizin-Routinedatenplattform in Germany[10,11]). Despite these efforts, broad access to medical records for research and development is not yet commonplace.

The lack of unified data systems not only slows advances in medical research but also negatively affects the health journeys of individuals. Incomplete medical records and difficulties accessing a patient's medical history compromise the overall quality of care that clinicians can provide. Furthermore, the effort required to compile, manage, and update personal medical information creates a barrier to engaging with one's health and limits the right of citizens to control their sensitive health data.

Several countries and health-care organisations are working on approaches for enabling individuals to access and manage their health data, but the implementation and maturity of such initiatives vary. In the USA, many health-care provider organisations offer access to patient portals with diverse features, but such systems are often not interoperable. For this reason, for-profit, third-party services collect and merge patients' health data on their behalf.[12] In Europe, health-care digitalisation efforts are coordinated on a national level but also differ substantially in their implementation. Approaches range from singular national databases as seen in Estonia,[7] to systems with multiple centralised organisations that store copies of the original data, as in Germany,[13] to decentralised systems, such as Elektronische Gesundheitsakte in Austria,[14] where data is stored by the health-care provider who first collected or generated the data. Although citizens can legally access their data in these countries, lingering issues regarding usability and real or perceived privacy concerns still prevent widespread adoption.[15,16]

Misunderstandings around privacy rights often hinder large-scale use or exploitation of health data outside of the immediate health-care delivery setting. The perceived conflict between the rights of citizens and the rights of potential data users aiming to improve research and care delivery for the common good creates a privacy-exploitation barrier that needs to be overcome to enable the best of both worlds.

To provide a general solution for overcoming this barrier within the heterogeneous digital health landscape in Europe, the European Commission recently proposed a draft for the European Health Data Space (EHDS) regulation.[17,18] This regulation aims to make electronic

health data widely available to EU citizens and their physicians. Furthermore, it outlines regulatory procedures for requesting access to anonymised health data for research, development, and regulatory purposes (figure 1).

Although this regulation might in principle overcome the privacy-exploitation barrier, we argue that a sustainable approach for building EHDS-compliant infrastructure still needs to be developed and resolved. Hasty and heterogeneous implementations at the member-state level will risk fragmenting the European data landscape and might lead to rejection by citizens and researchers alike.

Therefore, we argue that a common European system should not only enable interoperability across existing eHealth frameworks. Instead, it presents an opportunity to rethink national efforts to create a truly flexible and unified system that primarily focuses on meeting the needs of citizens and, furthermore, enables the use of health data for industry and research for decades to come. Therefore, we propose federated personal health data spaces, a truly citizen-centred solution for personal health data management in which citizens can set up and operate their personal health data spaces on digital devices controlled by them (figure 2), acting together as a federation to provide redundancy and technical resilience. Such a system would give citizens full control over their data and ensure that data concerning health stay with them wherever they go within Europe or beyond. By following a privacy-by-design approach, our proposed concept also aligns with the European General Data Protection Regulation (article 25: data protection by design and by default).[19]

See Online for appendix

We envision the proposed system as going well beyond the traditional doctor–patient relationship. Hence, we term this approach citizen-centred rather than patient-centred, because it encompasses the full spectrum of tools, ranging from digital wellness applications to wearables, fitness trackers, and preventive health care, and includes the acute treatment of citizens as patients.

With personal health data spaces, we propose a concept that would not only fulfil the requirements of the EHDS but go beyond them: personal health data spaces would give citizens full control over sharing their data with physicians, researchers, and others. Empowering citizens to shape their personal health data spaces according to their needs and preferences could help to enable more transparency, trust, and acceptance than existing national solutions. Science benefits as the willingness of citizens to participate in research increases through transparency and trust; as a result, previously infeasible large-scale study designs can emerge. We also provide a concept for using privacy-preserving federated algorithms that allow third parties to exploit medical data while protecting individual privacy. A more technical description of the concept and its connection to the proposed EHDS is provided in the appendix (pp 1–3).

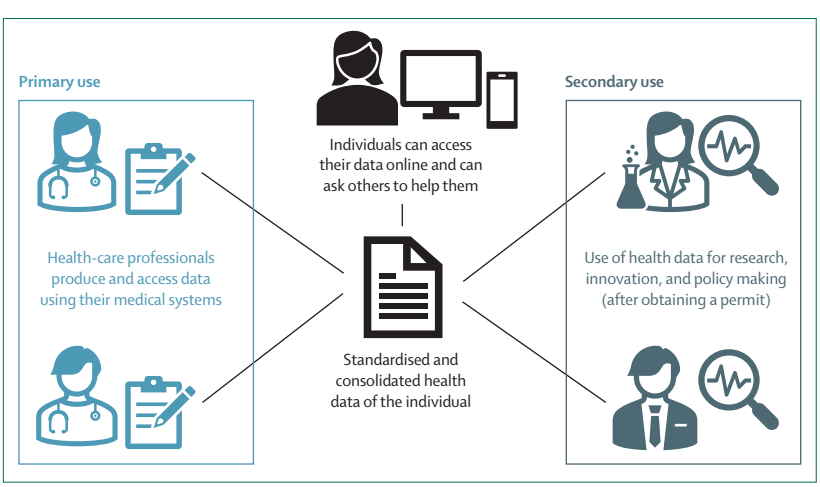

Figure 1: **Primary and secondary use of electronic health data as proposed by the EHDS**
The EHDS envisions that citizens can access, manage, and share their electronic health data in a standardised format. Primary use entails the sharing and processing of electronic health data with health-care professionals and related services for health care. Secondary use entails the sharing and processing of electronic health data for research, development, and regulatory purposes. Figure adapted from The European Commission Directorate-General for Health and Food Safety.[18] EHDS=European Health Data Space.

EHDS regulation

To implement compatible European eHealth systems, interoperability requirements are necessary to enable cross-border data exchange between health-care information technology systems and health-care professionals. The EHDS proposal[17] from May 3, 2022—which might still undergo minor changes—lays down these requirements for the EU in the context of two main use cases of electronic health data: primary use and secondary use (figure 1).

Primary use denotes the processing of electronic health data to support health care and related services. The introduction of national electronic health data access services enables citizens to view, manage, and share their electronic health data with health-care professionals who can access shared data through a separate national health professional access service. The EHDS proposal does not provide any specifications as to where the electronic health data should be stored. Instead, the two access services are intended to act as gateways. Citizens can enrich their electronic health record by adding real-world data from wellness apps and wearables, such as fitness trackers or smartwatches, as well as patient-reported outcomes collected while managing chronic diseases or after acute events or procedures. The proposal further requires data access to be logged, the ability to set up proxy access (for those who do not want to or cannot manage their health data independently), and enabling access by health-care professionals in emergencies.

Whereas primary use directly supports routine health care and health management, the EHDS defines secondary use as processing electronic health data for regulation and governance, education and scientific research, product and service development, algorithm

and AI development, and personalised health care. The EHDS explicitly forbids data processing for purposes that are harmful to citizens and for advertising or marketing. This restriction is enforced by providing access to non-identifiable health data for secondary use almost exclusively via national health data access bodies that review data access applications (appendix p 3). If a health data access body grants access, it collects relevant electronic health data from data holders (eg, health-care professionals, hospitals, or insurers) and provides a de-identified dataset to the applicant (the so-called data user) in a secure processing environment. The involvement of citizens in this process is not defined in detail and largely depends on legislation of member states, especially regarding important topics such as consent and citizen involvement.

Crucially, the EHDS does not provide technical implementation details. Without further clarification, the broad definitions and lack of clear guidance in the EHDS proposal will probably lead to the persistence of diverse national solutions (figure 3 shows how existing systems could be adapted in the context of the EHDS). This amplifies the risk that current issues and the opaque inner workings of some of these systems will persist, and the technical complexity of the existing national solutions will increase further. The EHDS proposal has the potential to fundamentally alter how citizens, health-care professionals, and data users interact with medical data. However, without a large-scale system overhaul, the wider aim of creating a seamless digital care process and advancing the next generation of data-driven medical research and development in Europe will not be met.

Federated personal health data spaces

The ongoing transition to patient-centred health care already helps to empower patients to participate actively in their health-care journey, leading to higher patient satisfaction and measurable improvements in health outcomes.[20,21] However, for health care to be truly citizen-centred, it is essential to also apply this approach of personal control and engagement to medical data management. Therefore, the primary goal of a reimagined health data management system should be to place the citizen at the centre of the emerging system—both conceptually and technologically. However, to uncritically hail this as the solution to all problems with data sharing and privacy would mean succumbing to technological solutionism,[22] compromising the rights of those who should be protected. Nevertheless, there is a need to identify and develop citizen-controlled solutions for those who are fundamentally dependent on better access to health data.

To support this citizen-centred health-care process, medical data solutions should provide citizens with full access to and control over their medical records, allowing for even deeper engagement with their health. Furthermore, we think that fundamental rights, such as access

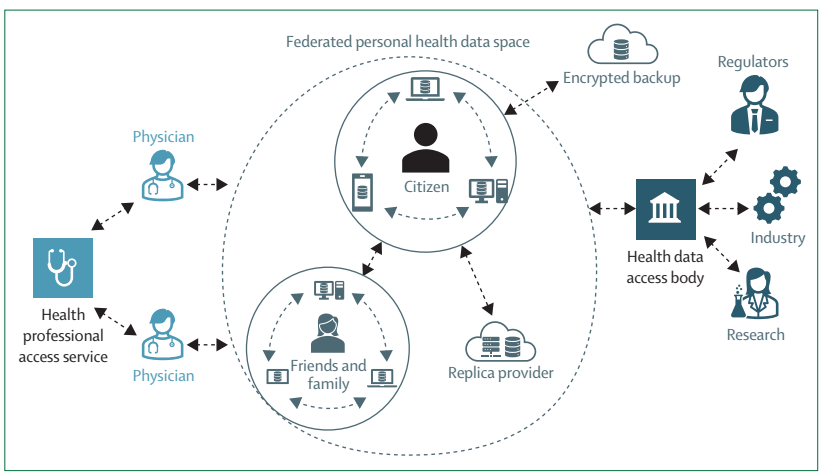

Figure 2: **Overview of the proposed federated personal health data space**
The concept of the personal health data space builds on a foundation of storing medical data on personal devices and constantly synchronising data across these devices. This approach is supported by encrypted backups to protect against data loss. In such a setup, physicians can look up a citizen's details through the national health professional access service and request data for primary use directly from each citizen's personal health data space. The national health data access body publishes secondary use requests, which are downloaded regularly by each personal health data space. If a citizen or their data match a request, the personal health data space software displays the request to the citizen; if they agree with the data sharing request, the personal health data space transmits the requested data to the health data access body (appendix pp 3–4).

to one's own personal information, should, to the extent possible, be guaranteed through the core design of a data management system and not merely through legal provisions.[23]

A personal health data space would incorporate the fundamental rights of citizens into its design by storing data on personal devices, such as smartphones, laptops, or computers, and synchronising data between them. Running the same synchronisation software on all devices and continuously keeping the data on each device up to date would allow the federation of devices to act as a unit that interacts with health-care professionals' computer systems and the national health data access body to enable primary and secondary use. Here, we outline the architecture and ideas underlying the personal health data space concept (figure 2). Further details on the implementation and integration into the EHDS are shown in the appendix (pp 1–3).

By giving citizens complete physical control over every aspect of their health data, a personal health data space would become private and safe, encouraging citizens to use it for storing and understanding their personal health and wellness data. Furthermore, we envision the personal health data space as a personal computing environment, allowing users to analyse private data on their own devices using certified algorithms to gain additional insights into their health journey without sharing data with a third party. This could enable novel AI-based telehealth approaches while eliminating some associated privacy concerns.

Patient access to electronic health records has been found to have positive effects, such as reassurance,

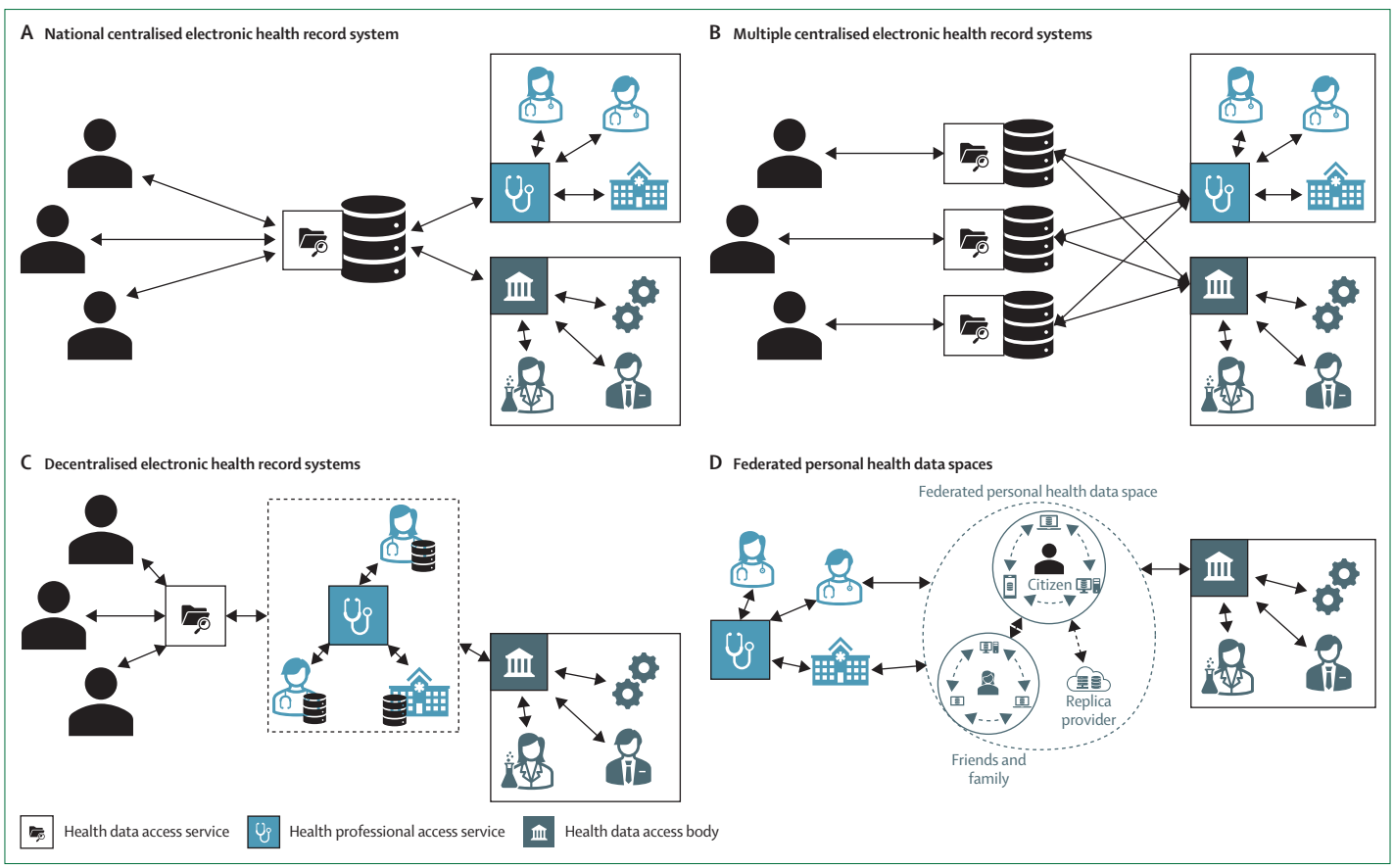

Figure 3: **Overview of possible implementation approaches within the EHDS framework**
Diagram of how existing eHealth frameworks could be implemented within the proposed EHDS. The EHDS foresees that individuals can access their personal health data through one or more health data access services. Health-care professionals access their patients' medical records through a central health professional access service. Data access for secondary use (research, development, and regulatory purposes) occurs through the national health data access body. (A) In a centralised national electronic health record system (as in Estonia), there is only one central storage system for all medical data, which can be accessed via the health data access service, health professional access service, and health data access body. (B) Patients access their data via a corresponding access service in a setting with multiple centralised electronic health record systems (as in Germany). Here, the health professional access service and health data access body provide access to all electronic health record systems. (C) In a system with multiple decentralised electronic health record systems (as in Austria), data are stored by health-care professionals who can connect to each other through the health professional access service. By contrast, the health data access service and health data access body provide access to all data pools. (D) In the proposed federated personal health data spaces, data are stored on individuals' devices, rendering the health data access service obsolete. Health-care professionals can look up the personal health data spaces of their patients through the health professional access service and directly connect to the relevant data space. The health data access body publishes lists of data requests to which personal health data spaces can react and share the requested data with the health data access body (if approved by the data space user). EHDS=European Health Data Space.

reduced anxiety, increased awareness of their overall health, and improved patient outcomes.[24] However, patients are concerned about the security, privacy, and confidentiality of their health data.[24] Personal health data spaces would require users to interact more actively with these topics, whereas existing systems place this obligation on system providers. Although the involvement of users would provide the chance for increased privacy and security, it could easily place an overwhelming burden on them. Against the backdrop of these concerns, a personal health data space could flexibly adapt to the requirements, perceived trust, and technical abilities of the users. Some citizens might decide to store data only on devices they physically own and control, to selectively share data only with doctors they trust, and to participate only in research they understand and believe to be worthwhile. Other citizens could decide to hand over control to a data fiduciary that is then in charge of managing their data, automatically accepting any data-sharing request from their physicians, and participating in any research study approved by the health data access body. In the latter case, personal devices would only be required as viewing devices for individual's data, similar to existing systems. This flexibility would allow the personal health data space to cater to the needs and requirements of more diverse groups of people than singular centralised solutions. It would also ensure that the personal health data space could act interchangeably with traditional solutions within the context of the EHDS proposal (appendix pp 1–2). Panel 1 provides an example of primary use interactions with personal health data spaces.

As well as empowering citizens and enabling primary use, personal health data spaces could also provide

Panel 1: Primary use process (eg, citizen–clinician interactions and data sharing)

In this hypothetical example of primary use interactions with personal health data spaces, Alex has recently moved and visits their new doctor, Robin, for a routine check-up. Before coming to the practice for the first time, Alex scheduled an appointment online and provided some form of digital identity. After receiving the appointment request, Robin's practice management system approves the appointment and recognises that Alex is a new patient and has not shared their medical data yet. The software system uses Alex's identity to request Alex's personal health data space connection details from a national registry and sends a data access request to Alex's personal health data space. Shortly after setting up the appointment, Alex receives a notification in the personal health data space application on their mobile phone, containing the data sharing request from Robin's practice management system. Alex can now decide what information to share with Robin. After approving the request, Robin's practice management system and Alex's personal health data space establish a bidirectional sharing connection until a time at which Alex revokes Robin's permissions (eg, after moving to a new doctor). Unless Alex chooses otherwise, all future data in Alex's personal health data space will be shared with Robin and all relevant data on Robin's side (eg, results of medical tests, diagnoses, and laboratory results) will be shared with Alex.

Panel 2: Secondary use process (eg, data sharing between citizen, health data access body, and researcher)

In this hypothetical example of secondary use interactions with personal health data spaces, Charlie is a researcher and wants to learn more about the connection between hypertension and specific medications. They send a data request to their national health data access body (according to the legal requirements of the European Health Data Space) specifying the required types of data and study inclusion criteria. The health data access body reviews the request and, after approval, publishes it in its register of open requests.

Luca has a personal health data space and is generally willing to share their data for research. However, they want to decide about data sharing on a case-by-case basis. Their personal health data space regularly downloads the list of new data requests from the health data access body register and locally checks whether the required data types are available and whether Luca meets published study inclusion criteria. In the case of Charlie's research, Luca's personal health data space contains the required data and Luca meets the study inclusion criteria. The personal health data space application therefore shows the request to Luca and asks for approval. Luca gets information about Charlie's background and affiliation, the planned data analysis, and the requested data categories. If Charlie has marked specific data categories as optional, Luca can exclude them from sharing. If Luca approves the request, the required data are shared directly with the health data access body.

The health data access body collects data from multiple personal health data spaces before anonymising and sharing the complete dataset with Charlie. Luca and all other participating citizens can choose to be notified about study updates and results provided by Charlie.

immense benefit for secondary use by allowing citizens to actively decide about data sharing for research and development purposes (panel 2). Although this might seem counterintuitive, studies have shown that people are more willing to share data if they receive information about the requesting party and can exercise control over the shared data categories.[25,26] Furthermore, the ability for researchers to interact with citizens would open up the possibility for large-scale population studies, which are infeasible with existing dominant recruiting methods. Beyond asking for consent, researchers could ask people to complete questionnaires, screen for active study participation, inform them about study outcomes, or ask them to enter or record data not yet available within their personal health data space (appendix p 2).

The personal health data space framework enables privacy-preserving methods for the secondary use of health data by providing local computational capabilities in addition to mere data storage. Tools such as differential privacy and federated learning do not require the sharing of identifiable personal health data with anyone—not even the health data access body (appendix pp 2–3)—but still enable the secondary use of health data. Differential privacy distorts data before sharing to minimise the risk of re-identification of citizens but has a minimal effect on the statistical analysis by the data users. Federated learning enables the distributed training of machine learning models on citizens' devices to support scientific discovery and the personalisation of AI models to individual requirements (panel 3).

In summary, the personal health data space framework empowers citizens by providing flexibility in handling their personal health data in three areas. First, the approach provides flexibility on a personal level regarding the structure of the individual personal health data space infrastructure. Here, the spectrum ranges from a self-hosted ecosystem managed exclusively by the citizen to an ecosystem hosted and managed by a data fiduciary. Second, the framework provides additional flexibility for decision making regarding data sharing for secondary use. Citizens can choose how to share information, whom to share information with, and how to make decisions about requests to share information. This flexibility ranges from an opt-out of all data sharing to a general opt-in to data sharing for any legitimate request. Intermediate options include making individual decisions about each request or agreeing in advance to share for specific purposes or with specific entities. Third, the framework provides flexibility regarding the data-sharing mechanism. Personal health data can be shared either directly without any modifications (for the health data access body to anonymise) or through privacy-preserving techniques such as differential privacy or federated learning, which do not share any identifying information.

This range of possibilities for citizens and data users provides the necessary tools for creating a transparent

Panel 3: Introduction to differential privacy and federated learning

Differential privacy is a formal privacy-preserving technique that enables the sharing of personal data without leaking information that could allow the identification or re-identification of individuals.[27] It distorts individual data values (such as diagnoses, measurements, or laboratory values) according to a specified probability distribution. Although individual datasets are too distorted to directly identify corresponding individuals, the collection of distorted data from many individuals can still be used to calculate useful statistics over the whole population of participants by removing the expected effect of the distortion process. Differential privacy provides theoretically sound guarantees on the amount of leaked privacy while enabling aggregated data analysis from multiple citizens.

Federated learning enables analysis—including the decentralised training of machine learning models—directly on the devices that contain the training data. In contrast to traditional machine learning, training data for federated learning do not need to be collected in a central location. Instead, the data stay on the original devices, and the training algorithm is sent to the data. A typical process works as follows. A central unit coordinates the training and distributes an initial and untrained version of the global model to all participating devices. All participating devices use their data to train the machine learning model for a few iterations before sending their model modifications back to the central coordinator. The coordinator collects and aggregates all model changes into an updated version of the global model. This process repeats until the model can perform the required task sufficiently. As federated learning on its own is not a fully privacy-preserving method, it must be combined with formal privacy techniques such as differential privacy or cryptography[28] to prevent adversaries from, for example, extracting sensitive data from the machine learning models.[29] When applied correctly, federated learning can open up large amounts of data for training machine learning models that would otherwise remain unused due to privacy concerns. In particular, when combined thoughtfully, differential privacy and federated learning can revolutionise data sharing by enabling privacy-preserving data analysis and artificial intelligence.

and trustworthy system that enables privacy for citizens and exploitation for data users—thus breaking the privacy-exploitation barrier that often stands in the way of medical data re-use.

Challenges for digital health ecosystems

Implementing any digital health solution is not without its challenges. Although centralised and federated systems each involve unique challenges and opportunities, they also share many. The following are some of the challenges and potential solutions within the personal health data space framework.

Data availability is an issue that both centralised and federated systems must reliably address to enable the availability of personal electronic health data for citizens, health-care professionals, and data users. Centralised systems become completely unavailable during system or network outages, whereas a user of a personal health data space can still access their health data, as long as they can access any participating personal device, and could even share data locally (eg, via a cable) with their physician. Replicating data across multiple devices within the personal health data space would provide resilience for the outage of single devices. Although temporary unavailability of data access is therefore unlikely in the context of a personal health data space, complete data loss could theoretically occur if all devices were to fail. However, the negative effect of data loss can be minimised by considering encrypted backups an essential part of the system (figure 2 and appendix p 1).

Beyond data availability, data security must also be maintained on an ongoing basis. Citizens who do not have the technical abilities to maintain the availability and security of their own personal health data space could choose data fiduciaries to provide a cloud-based replica of the space (figure 2). As with centralised system providers, these fiduciaries have the resources to address both data availability and security issues.

Data consistency and integrity are additional issues to be considered in the design of the personal health data space. They can be addressed through cryptographic methods that facilitate the keeping of a secure and tamper-proof audit log of all modifications, additions, and deletions in the personal health data space (appendix p 2).

Data quality always depends on the entities providing the data. In the envisioned system, data will be provided either by health-care professionals or by citizens themselves (through wearables, manual data entry, or digital health applications). Data quality cannot be enforced at a technical level (except for high-level consistency checks in structured data). However, cryptographic signatures could be used to ensure the traceability of the creator of each piece of data, which can incentivise health-care professionals to provide higher quality data and allow for future auditability if erroneous data are detected. Notably, the concepts of signatures and digital identities are not specific to the federated implementation and are also elementary components of centralised systems.

Standardised data formats and semantic and platform interoperability are necessary to enable the interconnection of the various stakeholders in the EHDS—both in a centralised approach and in the proposed federated solution. Work to establish common data formats and interoperability standards is ongoing at many levels and has yet to be fully resolved.

Finally, the introduction of any digital system—federated or centralised—for managing personal health data must be accompanied by educational initiatives targeting both digital and health literacy. Education is necessary to enable citizens to use the provided tools adequately and to be able to make informed decisions regarding data sharing and their personal health. Individuals need to be trained to consider the risks of each method, and what categories of data should be shared. The flexibility of the proposed personal health data space approach—especially the combination of data fiduciaries and proxy access (appendix p 1)—could bring benefits to citizens who might not be able or willing to access or operate digital devices by themselves.

Conclusion and outlook

Regulation and standardisation, as proposed by the EHDS, are necessary steps towards a harmonised, forward-looking digital health landscape supported by data-driven research. At the same time, the privacy of citizens should be preserved. The personal health data space concept presented here could not only support citizens' rights to privacy but also enable a range of new possibilities for secondary use of data and medical research and innovation.

Although the ideas presented here primarily address the requirements of the EHDS, they can be readily extended to other world regions. For example, individuals in the USA would benefit from being able to manage and control all their health data in one location, something that is currently infeasible, even given recent regulations designed to end information blocking.[30] Researchers worldwide would also benefit from access to much larger patient populations than would be accessible through individual health-care provider organisations.

Alongside the technical details of actual implementation, several open questions regarding user interactions remain to be solved. How should data requests be shown to individuals to allow them to make an informed decision? How can federated algorithms be explained to lead to a better understanding of the system? Moreover, a core question of any digital health system remains: how can digital and health literacy be improved? The proposed personal health data space concept provides a solid technical foundation for tackling these and other emerging questions for a future-proof, citizen-centred, innovation-oriented digital health ecosystem.

Contributors

RR conceived and drafted the initial manuscript. AK and AZ provided substantial writing, reviewing, and editing of the initial draft. All authors critically reviewed the manuscript and made edits to reach the final submitted and approved version.

Declaration of interests

AK declares funding from the Innovative Medicines Initiative 2 Joint Undertaking under grant agreement number 820820. ADS declares that she serves as a member of the Scientific Advisory Board of the German Society for Digital Medicine and the Strategic Advisory Board of HumanFirst. JK declares grants from the EU (for projects MobiliseD and DigiPD) and from the Bavarian Ministry of Economics (for project dHealthSolutions); payment for lectures from Roche/RoxHealth, Desitin, and Bial and for education from StreamedUP; support from Novartis for travel related to lecturing; a patent on gait assessment; a role as scientific advisor for Medical Valley Digital Health Application Center, Germany; membership in the patient-advisory board of EverPharma; an unpaid role in the European Taskforce for Digital Medical Devices; an unpaid advisory role in the Lux Taskforce for Health Technology Assessment for Digital Medical Devices; and has shares in Portabiles and Portabiles Healthcare Technologies. DR declares grants from the Alexander Humboldt Foundation, the European Research Council, the British Heart Foundation, Engineering and Physical Sciences Research Council, InnovateUK, Wellcome Trust, Horizon 2020, and German Federal Ministry of Education and Research; and a research contract with Roche. RE declares a grant from the Federal Ministry of Research and Education for the HiGHmed consortium in the Medical Informatics Initiative. BME declares consulting fees from Adidas, Siemens Healthineers, Portabiles Healthcare Technologies, and Sivantos; payment from AbbVie Deutschland and AGAPLESION; and a patent application EP 16174268.9 (gait assessment) with Portabiles HealthCare Technologies. All other authors declare no competing interests.

Acknowledgments

We thank Jürgen Geck, Leo Schwinn, and Matthias Braun for their feedback and comments. RR and AZ acknowledge funding provided by the Federal Ministry for Economic Affairs and Climate Action under grant number 68GX21004F (TEAM-X). JK acknowledges the PEARL dHealthPD support of the Luxembourg Research Foundation. GK acknowledges funding by the German Federal Ministry of Education and Research and the Bavarian State Ministry for Science and the Arts through the Munich Centre for Machine Learning and the Private Artificial Intelligence in Medicine (PrivateAIM) projects. SB acknowledges funding provided by Federal Ministry for Economic Affairs and Climate Action of Germany under grant number 68GX21001C (HEALTH-X). RE and HW acknowledge funding provided by Federal Ministry for Economic Affairs and Climate Action of Germany under grant number 68GX21001A (HEALTH-X). BME gratefully acknowledges the support of the German Research Foundation within the framework of the Heisenberg professorship programme (grant number ES 434/8-1). This work was supported by the Mobilise-D project that has received funding from the Innovative Medicines Initiative (IMI) 2 Joint Undertaking under grant agreement number 820820. This Joint Undertaking receives support from the European Union's Horizon 2020 research and innovation programme and the European Federation of Pharmaceutical Industries and Associations (EFPIA). Content in this publication reflects the authors' view and neither IMI nor the European Union, EFPIA, or any associated partners are responsible for any use that might be made of the information contained herein.

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